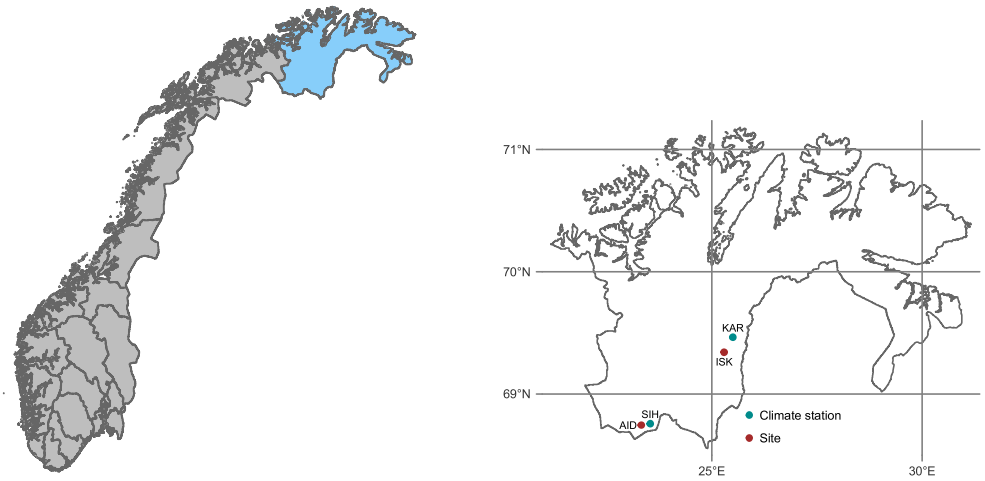
Chapter 2

Materials and methods

2.1 Site description

2.1.1 Iškoras



(a) **(b)**

Figure 2.1 – Shape of Norway (a), with Finnmark in blue, and locations of the sites Áidejávri and Iškoras as well as closest climate stations Karašjok and Šihččajávri (b).

Iškoras is located in Finnmark, northern Norway, at approximately 69.341 N and 25.296 E, ∼360 m above sea level (Fig. 2.1). The site is comparatively small, approximately spanning 5 ha (Naas et al., 2023). It shows severe signs of degradation, with large areas of thermokarst fens and ponds in between

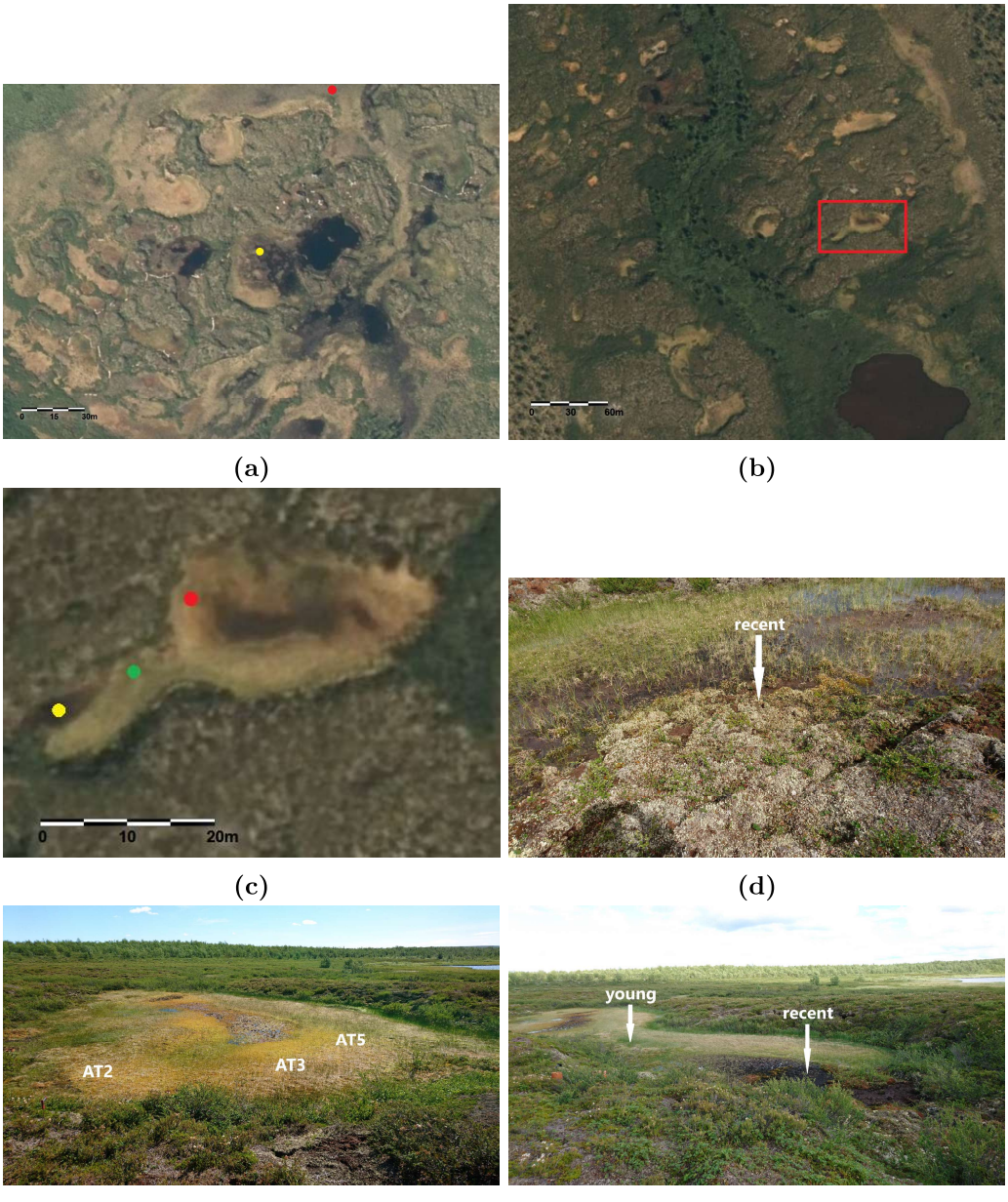
some remaining palsa complexes (Martin et al., 2019; Fig. 2.2 (a)). The scarce palsa vegetation is dominated by dwarf shrubs such as *Empetrum nigrum*, *Ledum palustre* and *Vaccinium uliginosum*, with some occurrence of *Vaccinium vitis-ideae*, *Betula nana* and *Andromeda polyfolia* as well as lichen (*Cladonia* spp.) and some mosses (see also Kjellman et al., 2018; Martin et al., 2019). Active layer depth in 2024 was around 60 cm by the end of the growing season. Thermokarst depressions are dominated by *Sphagnum* spp. and *Eriophorum* spp.. The depth of peat deposits is roughly between 1.5 and 2 m (Kjær et al., 2024; Kjellman et al., 2018; own findings). The age of the basal peat is around 9200 years and permafrost aggradation started approximately 800 years ago (Kjellman et al., 2018). It is assumed that the peatland developed through paludification, with no signs of limnic vegetation during peat initiation (Kjellman et al., 2018). Quartzite has been found to be the dominant bedrock type at Iškoras (Farbrot et al., 2013). The closest long-term recording weather station of the Norwegian Meteorological Institute, Karašjok-Markannjarga (131 m above sea level), reports a mean annual air temperature of −1.2 °C and a mean annual precipitation of 417 mm (1991 − 2020 reference period; Norwegian Center for Climate Services, n.d.).

2.1.2 Áidejávri

Áidejávri is located slightly south of Iškoras, approximately at 68.747 N and 23.327 E and ∼370 m above sea level (Fig. 2.1). It is much larger than Iškoras and, at time of sampling, comparatively large palsa complexes were still intact (Fig. 2.2 (b)). The vegetation is very similar to Iškoras. Peat deposits are around 2 m deep in the area used for sampling in this study (Fig. 2.2 (b, c)), but shallower in other parts (Kjær et al., 2024). Active layer depth was around 70 cm by the end of the growing season in 2024. Age of basal peat and permafrost aggradation are currently unknown (Kjær et al., 2024). The closest long-term recording weather station of the Norwegian Meteorological Institute, Šihččájavri (382 m above sea level), reports a mean annual air temperature of −1.6 °C and a mean annual precipitation of 577 mm (1991 − 2020 reference period; Norwegian Center for Climate Services, n.d.).

2.1.3 Selection of sampling locations

If not stated otherwise, all field work was carried out between mid-June and the beginning of September 2024. Three successional stages of thermokarst development were defined in a single thermokarst depression at Áidejávri (Fig. 2.2 (c)): One part of bare peat with almost no living vegetation was assumed to have thawed quite recently, without time for vegetation to es-



(a) (b) (c) (d) (e) (f)

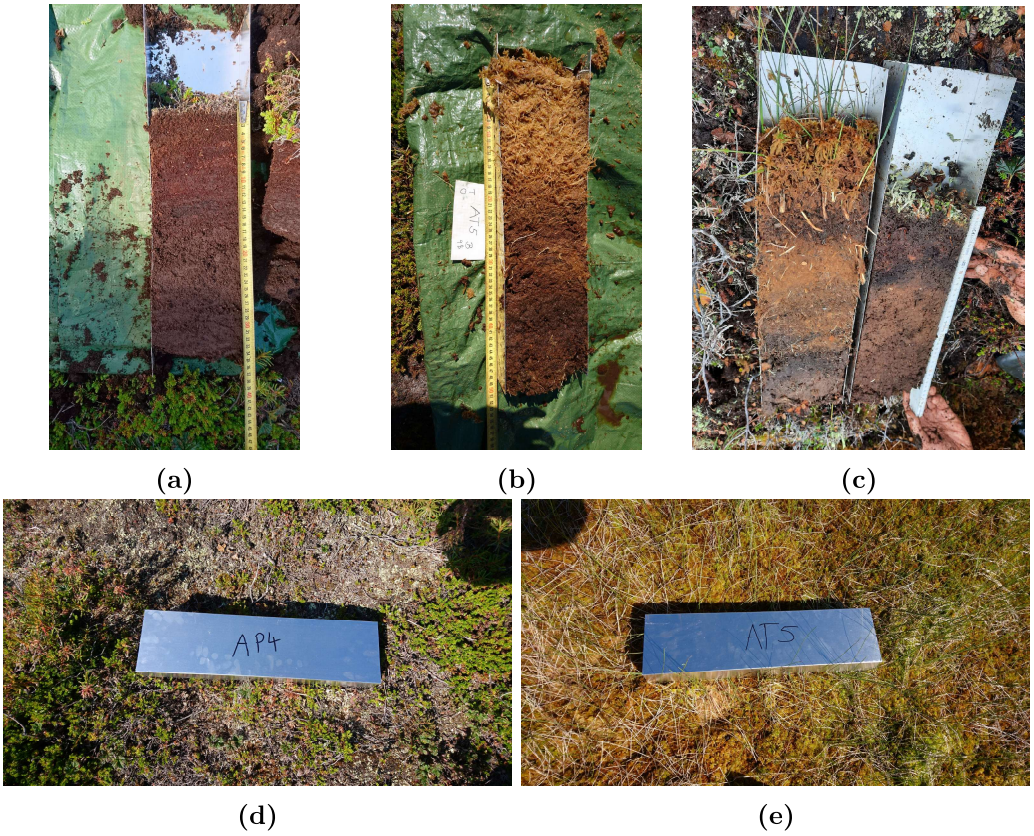
Figure 2.2 – Aerial images of Iškoras (a) and Áidejávri (b) as well as a close up of the investigated thermokarst depression at Áidejávri (c) from 2021 (Norwegian Mapping Authority, n.d.), with the locations of sampling marked in yellow (recent), red (mature) and green (young). Photos of the recent thermokarst at Iškoras (d), sampling locations of thermokarst cores in the mature thermokarst at Áidejávri (e) and locations of young and recent thermokarst at Aidejávri (f). Photos were taken 2024 during the field work for this study.

tablish (Fig. 2.2 (f)). Furthermore, two parts with apparently different dominant *Sphagnum* species could be identified: In one part, the *Sphagnum* cover was light-green while in the other part, its colour tended more towards orange (Fig. 2.2 (e, f)). Furthermore, the part with green *Sphagnum* had a higher coverage of *Eriophorum* sp. than the other part. The orange *Sphagnum* part was most likely dominated by *Sphagnum lindbergii* (see Fig. 3.7), a meso- to oligotrophic *Sphagnum* species that is characteristic for transitional stages between fen and bog (Gałka et al., 2017). The green *Sphagnum* species has not been determined yet, but could possibly be a more minerotrophic species like *Sphagnum riparium*, which would be indicative for a younger successional stage of thermokarst peatlands (Camill, 1999). Additionally, a higher cover of graminoids has also been associated with younger successional stages (e. g. Heffernan et al., 2022). An in-situ observed lower bulk density of the top peat in this part compared to the part dominated by *Sphagnum lindbergii* would additionally support the classification as a younger stage (Camill, 1999). Camill (1999) also suggest that more minerotrophic species usually occur closer to intact peat plateaus and show higher water tables than more oligotrophic species. At Áidejávri, both successional stages were within similar proximity to intact palsas of $\sim 2\,\mathrm{m}$. Mean water tables throughout the period of field work were also very similar in both parts and approximately 5 cm below the surface on average, while the mean water table in the recent thermokarst was directly at the surface. Also, the transition depth between pre- and post-thaw developed peat was, as far as observed in-situ, similar in presumably young and mature stages (roughly around 50 cm). Overall, since the species of the presumably young stage as well as the actual age of thaw have not been thoroughly determined yet, the classification of mature and young successional stages used in the following cannot be confirmed with certainty. Still, with the other criteria mentioned, the classification of young and mature stages seems to be likely valid.

At Iškoras, two successional stages were chosen, one with post-thaw accumulated peat of approximately $50-60\,\mathrm{cm}$. The *Sphagnum* species of the newly accumulated peat is currently unknown but the location generally appeared to be similar to the mature thermokarst at Áidejávri and is therefore also considered to be comparably mature. The other one, that had most likely collapsed very recently, still showed typical palsa vegetation while water saturated conditions prevailed (Fig. 2.2 (d)). Because of the palsa vegetation, it was assumed that this part had even thawed more recently than the recent thermokarst at Áidejávri. Contrary to Áidejávri, the two stages were located in different thermokarst depressions (Fig. 2.2 (a)). The mean water table was 11 cm and 6 cm below the surface in the mature and recent thermokarst, respectively.

The term "sampling locations" used in the following refers to different successional stages of thermokarst as well as palsa sampling locations at both sites. Abbreviations for sampling locations used in the following contain 2–3 letters, with the first letter referring to the site ("A" or "I"), followed by a letter to indicate palsa or thermokarst origin ("T" or "P") and, if all successional stages of thermokarst were included in measurements, a third letter indicating young, mature or recent ("Y", "M" and "R") thermokarst.

2.2 Peat core sampling



(a) **(b)** **(c)**

(d) **(e)**

Figure 2.3 – Soil core monoliths for AP4 (a), AT5 (b), and IT0/IP0 (c) as well as photos of vegetation at sampling locations for AP4 (d) and AT5 (e). For the rest of the monoliths, see Fig. A.2.

Cores from Áidejávri were sampled in June, July and August 2024. The three thermokarst cores analysed in this study (referred to as AT2, AT3 and AT5 in the following) were all taken from the mature area ($< 10\,\mathrm{m}$ between

core locations; Fig. 2.2 (e)). The top 50 cm were sampled with an aluminium tin (dimensions: 50*6*13 cm; Fig. 2.3; A.2) which was carefully inserted into the top soil by cutting with a bread knife around the edges, then hammered down until fully inserted. The aluminium tin containing the core was then lifted out manually. Deeper sections were sampled with a 50 cm Russian peat corer (diameter: 5.2 cm; Eijelkamp, Giesbeck, Netherlands). Two adjacent holes (approximately 30 to 50 cm apart) were used alternately, sampling each section with a 10 cm overlap to the preceding one until reaching mineral soil. The top 10 cm of each sampled section were later discarded to avoid contamination. The transition to mineral soil was identified visually. In the case of AT3, the mineral soil was not fully reached but it can be assumed that the tip of the Russian corer hit mineral ground and occasional sand grains were already visible in the deepest sampled section. Therefore, the missing part was most likely less than 10 cm and the impact on total carbon stocks was probably negligible. The transition depth between pre- and post-thaw developed peat was determined visually. It seemed obvious except for AT5, were it was uncertain if the transition depth was at 34 or 74 cm (Fig. 2.3 (b)). Therefore, in the following, both depths will be treated as potential transition depths.

Palsa cores at Áidejávri were sampled from surrounding palsas of the investigated thermokarst depression (AP2 and AP4) or approximately 100 m apart close to a different depression (AP1). In the palsas, the unfrozen active layer was sampled similarly to the thermokarst with the aluminium tin which was hammered into the soil until it hit frozen ground (active layer depth at time of sampling: 37 cm (AP2), 37 cm (AP4) and 39 cm (AP1); Fig. 2.3 (a); A.2 (a, b)). The frozen ground was sampled with a cylindrical steel pipe (3 cm diameter) which was hammered into the frozen ground in approximately 10 to 20 cm deep sections. Samples were pushed out of the steel pipe with a wooden dowel. A piece of cloth was pushed through the steel pipe in-between samples to avoid cross contamination. The length of each section was determined by measuring the depth inside the borehole since the length of the sample itself was not reliable due to expansions caused by varying ice contents of the permafrost samples. All sampled sections were wrapped into clean film, stabilised with half-cut PVC tubes for transport and frozen shortly after sampling. For subsampling, cores were unfrozen and cut with a knife and scissors into following sections: Monoliths were divided into a $4-6$ cm wide section to be used for chemical analysis and a 3 cm wide section for macrofossil analysis. For AT3 and AP1, the top 20 cm were cut at a 1 cm resolution. Deeper layers were cut at a $4-5$ cm resolution. For the rest of the cores from Áidejávri, monoliths were sampled at a 1 cm resolution and samples from greater depths at a 2 cm resolution. Samples taken from frozen

ground were measured and divided by the depth measured inside the borehole to determine the correct cutting intervals, assuming that the expansion within one sample was homogenous. Furthermore, those samples as well as samples taken with the Russian corer were cut in half lengthwise to use one half for chemical analyses and the other for macrofossil analysis. Macrofossils so far have been determined only for AT2 using the method described in Gałka et al. (2016).

At Iškoras, from palsas and thermokarst, one monolith each (IP0 and IT0) was sampled in September 2023 as described above (active layer depth for the palsa: 33 cm, length of the thermokarst monolith: 42 cm; Fig. 2.3 (c)). The thermokarst core was sampled from a location with approximately 30 cm of post-thaw developed peat. An 8 cm wide section of each monolith was cut at a 1 cm resolution, with the exception of IP0 below 20 cm depth, which was cut at a 2 cm resolution.

2.3 Chemical analysis of bulk peat

All samples were freeze dried and weighed afterwards to determine bulk density, with the exception of approximately every third sample from depths below 50 cm for AP1 and AT3, which was refrozen and stored. Samples from AP1 and AT3 were used to analyse organic matter (OM) content by loss on ignition (combustion of samples in a muffle furnace at 550 °C for 4 hours; Bhatti and Bauer, 2002; Chambers et al., 2011). For AP2, AP4, AT2 and AT5, one sample approximately every 10 cm was chosen for chemical analysis, leading to a total of 82 samples. From IT0 and IP0, 35 samples in total were chosen from selected depths. All chosen samples were ground to powder in a tungsten carbide ball mill (MM400, Retsch, Haan, Germany). For Fourier transform infrared spectroscopy (FTIR), 2 mg of powdered sample were mixed with 200 mg of KBr and pressed to 13 mm pellets. The pellets were then analysed for mid-infrared absorption (Cary 660, Agilent, Santa Clara, CA, USA). Absorption spectra were baseline corrected (method: "rubberband") and normalised using the R package "ir" (Teickner, 2022a). The ratio of the wavenumbers 1630/1090 was calculated as humification index using the R package "irpeat" (Teickner, 2022b). The index represents the ratio of aromatic compounds vs. polysaccharides in peat, with a higher value indicating stronger decomposition (Broder et al., 2012).

Elemental contents of phosphorus (P), sulphur (S), calcium (Ca), iron (Fe), copper (Cu) and manganese (Mn), among others, were analysed in cores from Áidejávri using wavelength-dispersive X-ray fluoresence spectroscopy (WD-XRF, Rigaku ZSK Primus II, Rigaku, Japan). 500 mg of powder were